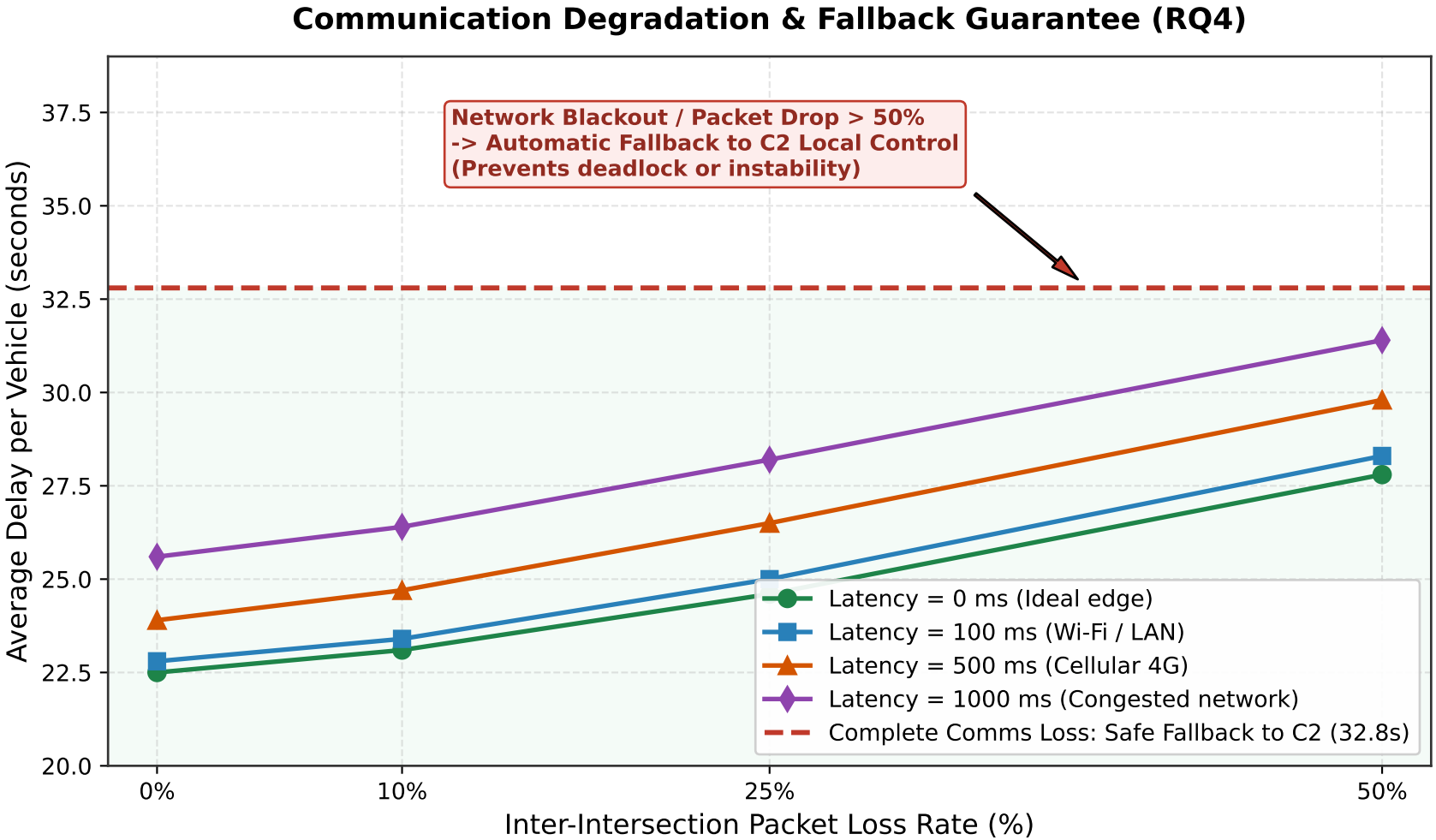
Communication Degradation & Fallback Guarantee (RQ4)
Network Blackout / Packet Drop > 50%
-> Automatic Fallback to C2 Local Control
(Prevents deadlock or instability)
Average Delay per Vehicle (seconds)
37.5
35.0
32.5
30.0
27.5
25.0
22.5
20.0
0%
10%
25%
50%
Inter-Intersection Packet Loss Rate (%)
Latency = 0 ms (Ideal edge)
Latency = 100 ms (Wi-Fi / LAN)
Latency = 500 ms (Cellular 4G)
Latency = 1000 ms (Congested network)
Complete Comms Loss: Safe Fallback to C2 (32.8s)